K-Means in Heart Disease

1. Read the file in SAS and display the contents using the PROC IMPORT and PROC PRINT procedures, print only the first 10 observations.

   SAS Code:

Editor - Untitled1 *

```
PROC IMPORT
OUT = heart_csv
datafile = "C:\Users\alshaban\Desktop\heart.csv"
dbms = csv replace;
getnames = yes;
proc print data= heart_csv (OBS =10);
    Title "First 10 Observations from the Heart CSV File";
proc contents data = heart_csv;
run;
```

   Results:

First 10 Observations from the Heart CSV File

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
2	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
3	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
4	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
5	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
6	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
7	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
8	44	1	1	120	263	0	1	173	0	0	2	0	3	1
9	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
10	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

First 10 Observations from the Heart CSV File

The CONTENTS Procedure

Data Set Name	WORK.HEART_CSV	Observations	303
Member Type	DATA	Variables	14
Engine	V9	Indexes	0
Created	2026-02-28 22:03:42	Observation Length	112
Last Modified	2026-02-28 22:03:42	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	WINDOWS_64		
Encoding	wlatin1 Western (Windows)		

Engine/Host Dependent Information	
Data Set Page Size	65536
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	584
Obs in First Data Page	303
Number of Data Set Repairs	0
ExtendObsCounter	YES
Filename	C:\Users\a1shaban\AppData\Local\Temp\SAS Temporary Files_TD34208_CCS-AVD-P5-3_\heart_csv.sas7bdat
Release Created	9.0401M5
Host Created	X64_SRV16
Owner Name	TMUAD\a1shaban
File Size	128KB
File Size (bytes)	131072

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
1	age	Num	8	BEST12.	BEST32.
12	ca	Num	8	BEST12.	BEST32.
5	chol	Num	8	BEST12.	BEST32.
3	cp	Num	8	BEST12.	BEST32.
9	exang	Num	8	BEST12.	BEST32.
6	fbs	Num	8	BEST12.	BEST32.
10	oldpeak	Num	8	BEST12.	BEST32.
7	restecg	Num	8	BEST12.	BEST32.
2	sex	Num	8	BEST12.	BEST32.
11	slope	Num	8	BEST12.	BEST32.
14	target	Num	8	BEST12.	BEST32.
13	thal	Num	8	BEST12.	BEST32.
8	thalach	Num	8	BEST12.	BEST32.
4	trestbps	Num	8	BEST12.	BEST32.

2. Perform basic Data analysis using PROC Means

SAS Code:

```
/* Basic Data Analysis Using PROC Mean */

proc means data= heart_csv;
run;
```

Output:

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
age	303	54.3663366	9.0821010	29.0000000	77.0000000
sex	303	0.6831683	0.4660108	0	1.0000000
cp	303	0.9669967	1.0320525	0	3.0000000
trestbps	303	131.6237624	17.5381428	94.0000000	200.0000000
chol	303	246.2640264	51.8307510	126.0000000	564.0000000
fbs	303	0.1485149	0.3561979	0	1.0000000
restecg	303	0.5280528	0.5258596	0	2.0000000
thalach	303	149.6468647	22.9051611	71.0000000	202.0000000
exang	303	0.3267327	0.4697945	0	1.0000000
oldpeak	303	1.0396040	1.1610750	0	6.2000000
slope	303	1.3993399	0.6162261	0	2.0000000
ca	303	0.7293729	1.0226064	0	4.0000000
thal	303	2.3135314	0.6122765	0	3.0000000
target	303	0.5445545	0.4988348	0	1.0000000

3. Apply standardization to the numerical attributes using stdize procedure and print the data (obs=10)

SAS Code:

Editor - Untitled1 *

```
proc means data= heart_csv;
run;

/* Standarization */
proc stdize method=std out=heart_csv_std;
    var age trestbps chol thalach oldpeak thal;
Proc print data=heart_csv_std (Obs=10);
Title "First 10 Observations after Standardization";
run;
```

Output - (Untitled) | Log - (Untitled) | Editor - Untitled1 * | Results Viewer - sashtm...

Output:

First 10 Observations after Standardization

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	-2.145323783	1
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	-0.512074772	1
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	-0.512074772	1
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	-0.512074772	1
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	-0.512074772	1
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	-2.145323783	1
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	-0.512074772	1
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	1.1211742386	1
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	1.1211742386	1
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	-0.512074772	1

4. Apply k-means clustering using fastclus procedure of SAS use your standardized dataset. Scatter plot your cluster labels (use y=chol and x=age) to visualize and compare with the original data labels. Assuming that you do not know the exact number of clusters in the dataset, try k=2, 3, 4, 5 and evaluate the solutions. Choose the best K value based on the RMS Std.Deviation

<u>SAS Code:</u>

Editor - Untitled1 *

```
/*K-means Clustering*/

%macro doFASTCLUS;
    %do k=2 %to 5;
            title
                    'Clusters in Heart with' &k'=Means';
            proc fastclus
                data= heart_csv_std
                out = clust_solution
                maxiter= 100
                maxclusters =&k
                summary;

            proc sgplot;
                scatter x=age y=chol / datalabel =cluster;
            run;
    %end;
%mend;
%doFASTCLUS

proc print data=clust_solution (Obs=10);
run;
```

The solution code is first standardizing the data because the scale of variables is not the same. The stdize procedure creates the output dataset heart_csv_std that contains standardized data (mean 0 standard deviation 1) for specified variables.

Next, the fastclus procedure is applied in a loop to find clusters with k=2, 3, 4, 5. It takes heart_csv_std as input and outputs a clust_solution.

For k=2:

Cluster Summary:

Results Viewer - sashtml.htm

Clusters in Heart with k=2-Means

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=2 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.7491

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	178	0.7098	6.7719		2	2.6452
2	125	0.8071	5.0672		1	2.6452

Pseudo F Statistic = 65.00

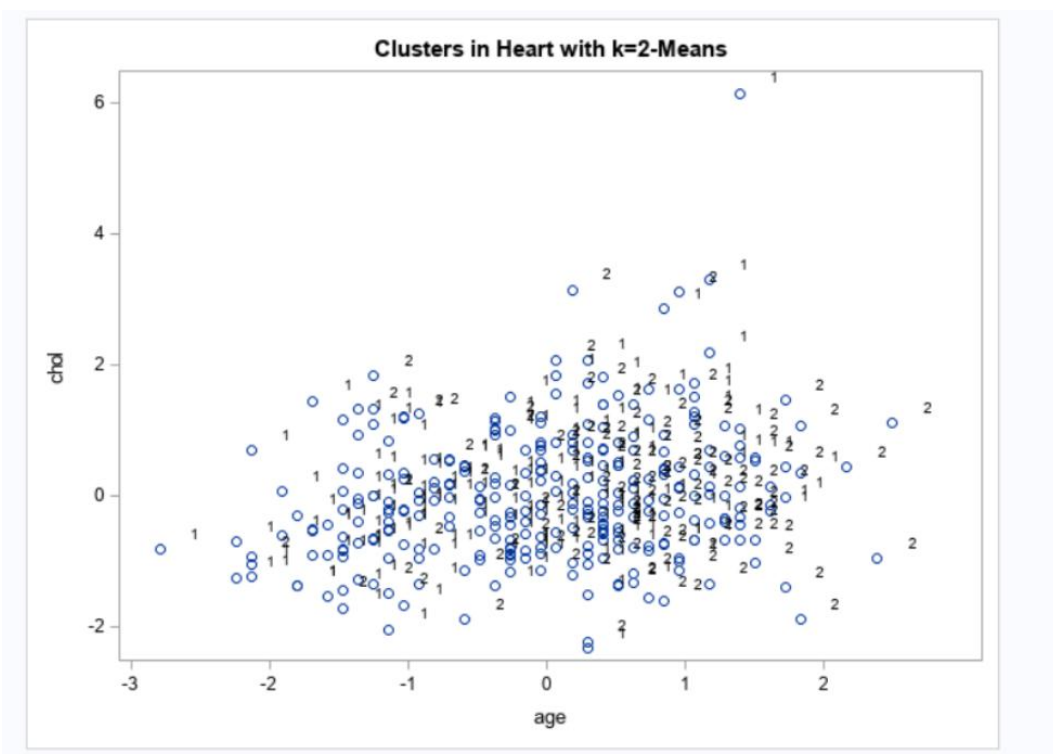
Observed Over-All R-Squared = 0.17759

Approximate Expected Over-All R-Squared = 0.08887

Cubic Clustering Criterion = 22.716

WARNING: The two values above are invalid for correlated variables.

Scatter Plot:



First 10 Observations:

Clusters in Heart with k=2-Means

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	CLUSTER	DISTANCE
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	-2.145323783	1	1	3.91108
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	-0.512074772	1	1	3.72045
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	-0.512074772	1	1	1.97952
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	-0.512074772	1	1	1.43948
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	-0.512074772	1	1	2.96005
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	-2.145323783	1	1	2.94177
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	-0.512074772	1	1	1.90256
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	1.1211742386	1	1	2.03426
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	1.1211742386	1	1	3.20768
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	-0.512074772	1	1	2.51017

For k=3:

Cluster Summary:

Results Viewer - sashtml.htm

Clusters in Heart with k=3-Means

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=3 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.7143

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	137	0.6679	4.3421		3	2.2306
2	99	0.7697	4.9384		3	2.7228
3	67	0.7361	6.1446		1	2.2306

Pseudo F Statistic = 50.65

Observed Over-All R-Squared = 0.25243

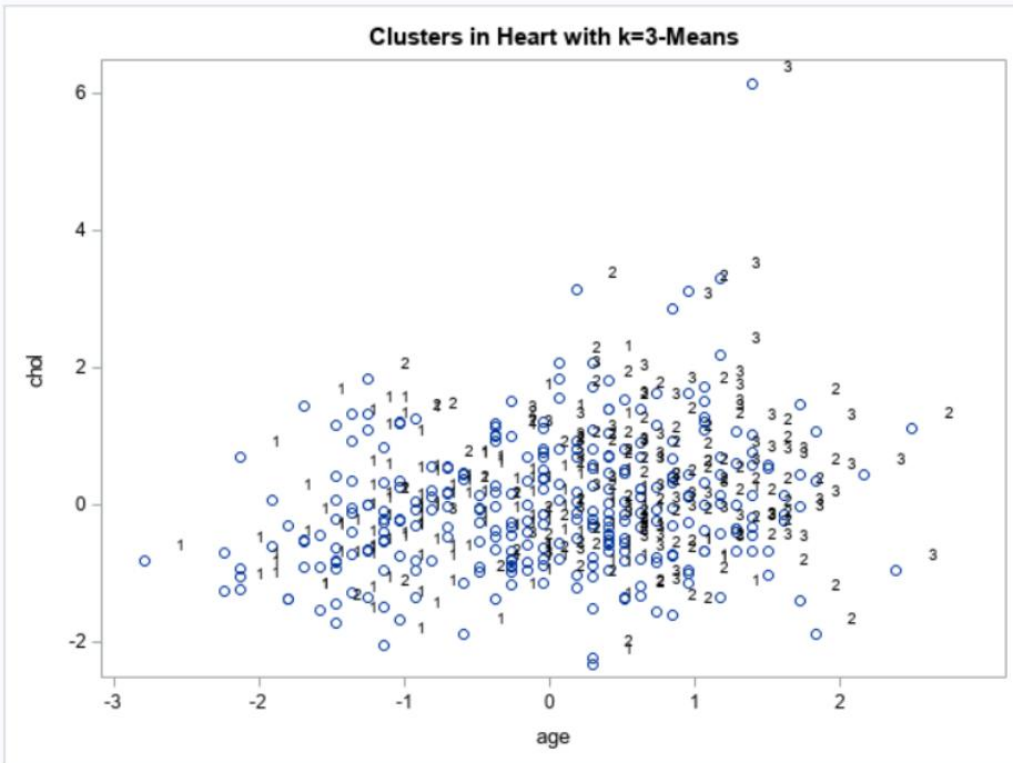
Approximate Expected Over-All R-Squared = 0.15695

Cubic Clustering Criterion = 19.161

WARNING: The two values above are invalid for correlated variables.

Output - (Untitled) | Log - (Untitled) | Editor - Untitled1 * | Results Viewer - sasht...

Scatter Plot:



First 10 Observations:

Clusters in Heart with k=3-Means

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	CLUSTER	DISTANCE
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	-2.145323783	1	3	3.23362
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	-0.512074772	1	1	3.65526
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	-0.512074772	1	1	1.73235
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	-0.512074772	1	1	1.43212
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	-0.512074772	1	1	3.12042
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	-2.145323783	1	1	2.85330
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	-0.512074772	1	3	1.56970
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	1.1211742386	1	1	1.91775
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	1.1211742386	1	3	3.13349
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	-0.512074772	1	1	2.66314

For k=4:

Cluster Summary:

Results Viewer - sashtml.htm

Clusters in Heart with k=4-Means

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=4 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.6979

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	30	0.6897	5.1688		2	2.4471
2	54	0.7144	4.1667		3	2.3632
3	122	0.6419	4.3388		2	2.3632
4	97	0.7675	4.9979		2	2.8003

Pseudo F Statistic = 40.22

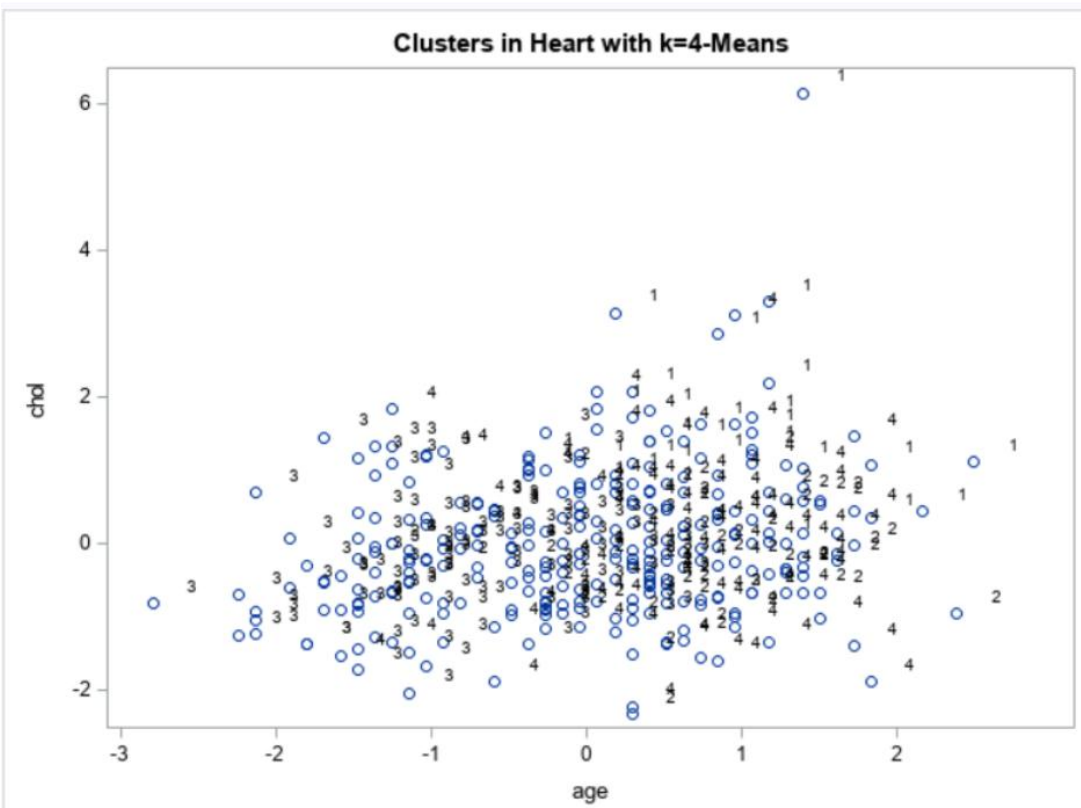
Observed Over-All R-Squared = 0.28751

Approximate Expected Over-All R-Squared = 0.21034

Cubic Clustering Criterion = 14.156

WARNING: The two values above are invalid for correlated variables.

Scatter Plot:



First 10 Observations:

Clusters in Heart with k=4-Means

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	CLUSTER	DISTANCE
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	-2.145323783	1	2	3.01633
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	-0.512074772	1	3	3.60130
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	-0.512074772	1	3	1.69662
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	-0.512074772	1	3	1.43908
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	-0.512074772	1	1	1.96931
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	-2.145323783	1	3	2.88078
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	-0.512074772	1	1	1.75271
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	1.1211742386	1	3	1.88969
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	1.1211742386	1	2	2.70472
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	-0.512074772	1	2	2.10265

For k=5:

Cluster Summary:

Results Viewer - sashtml.htm

Clusters in Heart with k=5-Means

The FASTCLUS Procedure
Replace=FULL Radius=0 Maxclusters=5 Maxiter=100 Converge=0.02

Convergence criterion is satisfied.

Criterion Based on Final Seeds = 0.6804

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	103	0.6262	4.3504		4	2.2003
2	45	0.7437	4.5521		3	2.0738
3	68	0.7412	4.7432		2	2.0738
4	62	0.6888	4.1956		1	2.2003
5	25	0.6495	5.2699		4	2.3634

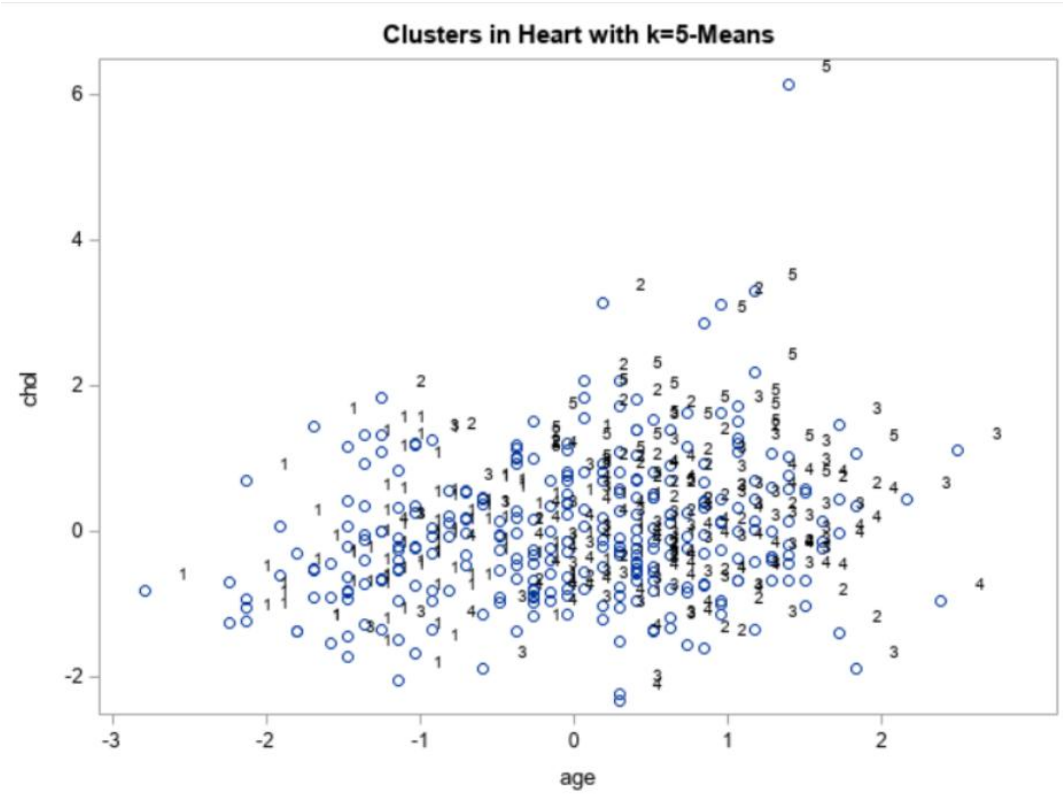
Pseudo F Statistic = 35.32

Observed Over-All R-Squared = 0.32161

Approximate Expected Over-All R-Squared = 0.25577

Cubic Clustering Criterion = 11.656

Scatter Plot:



First 10 Observations:

Clusters in Heart with k=5-Means

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target	CLUSTER	DISTANCE
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	-2.145323783	1	4	2.95011
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	-0.512074772	1	1	3.57558
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	-0.512074772	1	1	1.65887
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	-0.512074772	1	1	1.51681
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	-0.512074772	1	5	2.04373
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	-2.145323783	1	1	2.98626
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	-0.512074772	1	5	1.62575
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	1.1211742386	1	1	1.75673
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	1.1211742386	1	4	2.90950
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	-0.512074772	1	4	2.05252

Conclusion:

Ideally, an appropriate value of k should maximize the separation and minimize the cohesion. Comparing all solutions reveals that **k=4** provides optimal separation, cohesion, and homogeneity.